# SAI ARUN GUNUGANTI

**Senior Software Engineer · Lead Developer | Frontend Platform, Micro-Frontends & Agentic Systems**

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## SUMMARY

Senior Software Engineer and Lead Developer with 10+ years building large-scale web platforms for retail, banking, and healthcare (Lowe's, Apple, Cisco, McKesson, M&T Bank). At Lowe's I lead a team of 5 and own the checkout-critical Customer Lookup micro-frontend, built from scratch and integrated by 20+ teams, while contributing across 25 repositories spanning checkout, quality engineering, observability, and AI agents (Lowe's serves ~16M customer transactions a week across 1,700+ stores). Outside work I build agentic systems, including a live AI-agent authorization product with a published technical paper.

## CORE SKILLS

- **Frontend & Platform:** TypeScript, JavaScript, React, Redux / Redux-Saga, React Query, Vue.js, Angular, Webpack Module Federation (Micro-Frontends), Electron, design systems
- **Quality & Automation:** Playwright (Electron), Cypress, Jest, React Testing Library, Mock Service Worker, contract assertions, Allure, CI quality gates
- **Backend, Identity & Delivery:** Node.js, Python, FastAPI, BFF services (Micronaut / Spring Boot), REST, ForgeRock / Keycloak, GCP, Jenkins, Azure DevOps, GitOps
- **AI / Agentic:** Multi-agent architecture, LLM integration and chat routing, agent trace evaluation, MCP, Google agent SDK
- **Observability & Leadership:** Telemetry SDKs, FullStory, ELK, Grafana · technical leadership, mentoring, code review, design docs, engineering standards

## PROFESSIONAL EXPERIENCE

**Lowe's Companies Inc. — Senior Software Engineer, Lead Developer (myRedVest Showroom POS)** Apr 2022 – Present

- **Customer Lookup Micro-Frontend:** built it from scratch (Webpack Module Federation, custom events, React Query); it is **integrated by 20+ teams** into their own applications. Developed customer-profile, quote/estimate recall, preferred-contact, and address-validation workflows, with ForgeRock/Keycloak authentication, continuation through API failures, and performance and error telemetry.
- **Technical leadership:** Lead Developer for a team of 5 (3 onshore, 2 offshore) and single point of contact for a core area of checkout. I mentor engineers, own code review, and wrote the design docs and integration conventions (event contracts, data requirements) that consuming teams follow; I onboarded those teams onto the MFE.
- **Cross-MFE checkout integration:** integrated customer lookup, cart, pre-tender, tender, and customer-facing display micro-frontends in the React/Redux POS shell, coordinating event contracts, checkout state, authentication, and fulfillment. Built synchronized associate/customer dual-monitor flows (protection plans, installation, product discovery) over bidirectional platform events.
- **Checkout features:** installation appointment scheduling, boom-truck delivery fees and eligibility, outlet-store rules, mixed-material and split-delivery carts, subscription-aware tender restrictions, and Home Care+ in the TypeScript POS shell. Improved Pro quoting reliability.
- **Test automation:** engineered the Playwright/Electron automation framework for retail selling flows, with reusable screen and business-flow abstractions, Stage/PROD runtime lanes, contract assertions, Linux and Windows runners, and Allure/email reporting. Run in UAT ahead of production deploys, it **caught critical failures in key POS transaction flows before release**; I escalated them to the owning teams so catastrophic checkout failures were fixed before reaching stores.
- **Change-driven quality platform:** built a platform that maps repository changes to business capabilities and flow specs, generates Playwright test candidates, validates them against policy and evidence, and supports governed CI authoring and promotion.
- **Observability:** built shared TypeScript telemetry utilities for micro-frontends, covering normalized events and action spans, privacy and sampling controls, bounded transport delivery, and cross-bundle action coordination.

- **AI / agentic systems:** designed an agentic architecture for POS guided selling and prototyped a Showroom assistant with domain-specialized agents, LLM integration, unified chat routing, and repository-grounded troubleshooting. Built a Python/FastAPI service that evaluates agent trace evidence against capability contracts, with separate Stage/PROD verdicts.
- **Production ownership:** first point of contact for production issues and change coordinator through release (CRQs, deployment validation, CAB approvals up to Director/CIO level). Maintained GitOps manifests for seven POS workloads and remediated dependency and payload-handling vulnerabilities across shared logging, POS, and checkout repositories.

**Atos Syntel Inc. — Consultant** Nov 2020 – Apr 2022

- **McKesson:** Vue.js front-end (single-file components, Vuex, Element UI) and Okta Verify fixes. **Baylor Scott & White Health:** React apps with React Query and Mock Service Worker, and reusable components for the bswhealth.com Sitecore migration.

**SLK America Inc. — Senior Software Engineer** Sep 2018 – Nov 2020

- **M&T Bank:** React digital banking (e-statements, EZ-Money P2P, Alexa banking, feature-flag dashboard), WCAG/ADA accessibility, and the Git flow for the TFS to Azure DevOps migration. **First Horizon Bank:** Angular 6 / TypeScript loan-operations BPM tool with reusable component libraries.

**Valley Soft Systems Inc. — Software Engineer** Sep 2017 – Aug 2018

- **Apple (Talent Point):** full-stack Meteor / React / Node.js / MongoDB on the careers platform. **Cisco (DNA Center):** React/Redux product UI and the shared Cisco-DNA-react-components library.

**Alliance Information Technology Inc. — Software Engineer / Front-End Developer** Aug 2016 – Sep 2017

- **Symantec (Rev-Pro):** React/TypeScript executive financial dashboard for C-level reporting; earlier JavaScript/jQuery vendor dashboards in India (Jun 2014 – Aug 2015).

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- **Sealgraf** (live product, sealgraf.com): designed and built the authorization layer for AI-agent spending, with owner-set limits checked before money moves, approval on a surface the agent cannot fake, and sealed, verifiable receipts. Built integration packages for **6 agent frameworks** (Dify, CrewAI, LangChain, Vercel AI SDK, n8n, Composio; 302 tests) and authored the technical paper *"Approval You Can Verify"*.
- **Eight sector agents** (live public demos; 378 tests): Doorway, ListProof, Dossier, JobProof, Reclaim, Refile, Destorm, and ModSeal. All run on one spine: a state-machine orchestrator, rules in code, approval gates, and a hash-chained receipt ledger.
- **Reify** (open source, github.com/saiarung27/reify): a local-first memory engine for AI agents with a bitemporal graph, hybrid retrieval, and MCP / REST / CLI surfaces. 442 tests

## EDUCATION

M.S., Computer Information Systems & IT, University of Central Missouri (2016) · B.Tech, Mechanical Engineering, Anurag Engineering College (2014)